

BIKIN, Russian Federation

WMO: 318320

Lat: 46.800N

Lon: 134.267E

Elev: 221

StdP: 14.58

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.3	-21.3	-29.3	1.1	-24.9	-25.7	1.3	-20.5	12.8	8.1	11.3	7.5	0.9	100	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	85.9	71.7	83.5	70.8	80.8	69.2	74.7	82.4	73.2	80.4	71.6	78.2	4.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	121.0	79.3	70.7	114.2	77.6	69.1	108.0	76.2	38.6	82.5	37.1	80.6	35.7	78.2	80.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.2	9.6	DB	-29.9	90.8	4.9	2.2	-33.4	92.4	-36.3	93.7	-39.1	95.0	-42.6	96.6	(4)
(5)			WB	-30.1	77.3	4.9	1.4	-33.6	78.3	-36.4	79.1	-39.2	79.9	-42.7	81.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	-3.3	5.7	23.3	42.2	56.3	65.7	72.2	69.4	57.9	42.7	21.0	1.6		
	DBStd	27.54	8.51	9.36	10.53	8.55	7.12	6.13	4.35	4.86	6.90	8.29	11.62	9.91		
	HDD50	6648	1651	1241	828	266	21	0	0	0	13	257	871	1500		
	HDD65	10330	2116	1661	1292	686	282	64	4	17	229	693	1321	1965		
	CDD50	2287	0	0	0	31	217	471	689	601	249	29	0	0		
	CDD65	496	0	0	0	1	13	85	229	153	15	0	0	0		
	CDH74	3209	0	0	0	15	171	608	1500	828	86	1	0	0		
	CDH80	708	0	0	0	2	26	148	388	137	7	0	0	0		
	WSAvg	3.9	2.9	3.3	4.1	5.0	4.9	4.2	3.8	3.7	3.7	4.1	4.2	3.2		
	PrecAvg	25.4	0.5	0.4	0.7	1.6	2.4	3.5	4.4	4.9	3.5	2.1	1.1	0.7		
(15) Precipitation	PrecMax	33.3	1.1	1.1	2.7	4.1	5.6	10.2	8.3	10.9	6.5	6.1	3.2	2.2		
	PrecMin	15.1	0.0	0.0	0.2	0.4	0.6	1.1	0.6	0.4	0.6	0.0	0.2	0.1		
	PrecStd	4.2	0.3	0.3	0.5	0.9	1.0	1.7	2.1	2.2	1.3	1.5	0.6	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.3	37.5	53.0	75.5	83.4	88.5	89.3	86.5	80.7	70.1	53.2	35.4		
		MCWB	23.1	32.6	41.5	55.5	63.8	70.5	74.5	73.3	67.7	56.9	46.8	31.6		
	2%	DB	19.5	32.6	47.4	68.4	78.9	84.2	86.6	83.7	76.5	65.0	45.8	28.3		
		MCWB	17.1	28.2	37.9	52.3	61.3	68.9	72.9	72.0	65.4	53.6	39.7	26.4		
	5%	DB	14.7	26.7	43.4	62.4	74.4	80.6	84.4	81.0	73.1	60.4	41.8	22.5		
		MCWB	12.8	23.2	35.3	48.6	58.6	66.9	72.0	70.5	63.1	51.2	36.8	20.5		
	10%	DB	10.8	21.9	39.4	57.9	70.3	77.1	81.7	78.6	70.0	56.6	37.4	17.0		
		MCWB	9.2	18.9	33.1	45.8	56.5	65.7	71.1	69.5	61.3	48.6	33.7	15.2		
	0.4%	WB	23.1	33.7	42.8	56.7	65.6	74.5	77.4	75.8	69.9	59.1	46.1	32.5		
		MCDB	25.0	36.1	49.3	73.1	80.5	84.9	85.5	82.8	77.3	67.4	54.0	35.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	17.1	28.1	38.8	52.5	62.6	71.1	75.5	74.0	67.1	55.1	40.2	26.7		
		MCDB	19.3	32.5	46.3	65.9	75.9	80.7	83.4	80.8	74.0	61.9	44.8	28.2		
	5%	WB	12.9	23.3	36.1	49.5	60.2	68.9	74.0	72.5	64.8	52.7	37.0	20.6		
		MCDB	14.6	26.9	41.9	60.8	71.8	77.7	81.7	78.8	71.0	59.4	41.2	22.4		
	10%	WB	9.3	18.9	33.5	46.5	57.8	66.9	72.5	71.0	62.6	49.4	34.0	15.4		
		MCDB	10.9	21.8	39.2	56.4	67.7	75.0	79.5	76.8	68.4	55.6	37.0	17.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.5	24.3	20.8	20.6	19.8	17.1	15.4	15.3	19.7	19.3	16.9	18.7		
		MCDBR	23.4	23.4	21.8	28.6	27.2	22.4	20.5	18.8	22.8	25.0	19.1	19.3		
		MCWBR	21.4	19.9	14.0	15.8	13.5	10.1	8.8	8.8	12.1	15.0	14.0	17.9		
		MCDBR	22.8	22.4	19.6	26.3	23.5	19.4	17.2	15.1	18.9	21.4	17.0	19.1		
	5% WB	MCWBR	20.9	19.2	13.3	15.0	12.5	9.6	8.6	7.9	10.9	13.7	13.1	18.0		
(40) Clear-Sky Solar Irradiance	taub		0.303	0.346	0.428	0.537	0.570	0.536	0.510	0.468	0.422	0.440	0.384	0.320		
	taud		2.089	1.953	1.811	1.746	1.751	1.928	2.051	2.168	2.203	1.990	2.029	2.056		
	Ebn at Noon		239	249	244	229	227	237	242	247	247	213	200	212		
	Edh at Noon		30	42	57	67	69	58	51	44	38	41	32	28		
(44) All-Sky Solar Radiation	RadAvg		636	1008	1377	1449	1643	1758	1636	1461	1241	845	562	486		
	RadStd		38	47	109	146	145	215	150	93	109	64	48	33		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page